



Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Parallel Computing–I part–Thursday, June 08th, 2023

Polimi ID _____

Surname _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 3 (4 points) _____

Exercise n. 1

Answer the following questions about PRAM models and briefly explain (without an explanation, the answer will be considered invalid)

A. Describe the steps followed by a PRAM machine (1)

B. All processors in a PRAM model can have non-uniform memory access. True or False? (1)

C. Complete the following sentence and explain what it means: "A **write conflict** occurs when..." (1)

D. Amdahl's model is a fixed-time model. True/False? (1)

Exercise n. 2

Answer the following questions about different forms of parallel executions and briefly explain (without an explanation, the answer will be considered invalid).

- A. Please summarize the main differences between SIMD and Multi-core. (1)

- B. Define what a memory stall is and why caches reduce the length of stalls. (1)

- C. Explain why "**A processor with multiple hardware threads has the ability to *avoid stalls.***" (1)

- D. Write a short description of "**Simultaneous multi-threading.**" (1)

Exercise n. 3

Answer the following questions about programming models plus CUDA and briefly explain (without an explanation, the answer will be considered invalid).

A. Is CUDA a data-parallel programming mode? (1)

B. How scalar registers for CUDA thread are organized in an NVIDIA V100 SM "sub-core"? (1)

C. Describe the distinct types of address spaces visible to kernels. How many? How much is shared? How faster are they? (2)

Exercise n. 4

Answer the following questions about memory and heterogeneous systems and briefly explain (without an explanation, the answer will be considered invalid).

A. Please briefly describe what DRAM burst means. (1)

B. Provide an example of CUDA un-coalesced accesses? (1)

C. Please explain what a Total Store Ordering consistency model does. (1)

D. What is meant by heterogeneous processing? (1)
